

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

WINTER SEMESTER, 2022-2023
FULL MARKS: 150

CSE 4711: Artificial Intelligence

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. Consider placing a robot in an unfamiliar environment. The robot perceives its current state, s_t , takes an action, a_t , transitions to a new state, s_{t+1} , and receives an instant reward, r_t . Assuming a discount factor of 0.5 and a learning rate of 0.5, the robot explores the environment, resulting in the experiences outlined in Table 1.

Table 1: Experience achieved in 5 iterations for Question 1

t	s_t	a_t	s_{t+1}	r_t
0	<i>X</i>	<i>Left</i>	<i>Y</i>	2
1	<i>Y</i>	<i>Left</i>	<i>Y</i>	-4
2	<i>Y</i>	<i>Right</i>	<i>Y</i>	0
3	<i>Y</i>	<i>Right</i>	<i>X</i>	3
4	<i>X</i>	<i>Right</i>	<i>X</i>	-1

- a) Formulate the scenario as a sample-based Q-learning problem by determining Q-values for each state-action pair based on the experience of the robot.

24
(CO2)
(PO2)

Solution:

At $t = 0$, we get:

$$\begin{aligned} Q(X, Left) &= (1 - \alpha)Q(X, Left) + \alpha(r + \gamma \max_{a'} Q(Y, a')) \\ &= (1 - 0.5) \times 0 + 0.5 \times (2 + 0.5 \times 0) \\ &= 1 \end{aligned}$$

At $t = 1$, we get:

$$\begin{aligned} Q(Y, Left) &= (1 - \alpha)Q(Y, Left) + \alpha(r + \gamma \max_{a'} Q(Y, a')) \\ &= (1 - 0.5) \times 0 + 0.5 \times (-4 + 0.5 \times 0) \\ &= -2 \end{aligned}$$

At $t = 2$, we get:

$$\begin{aligned} Q(Y, Right) &= (1 - \alpha)Q(Y, Right) + \alpha(r + \gamma \max_{a'} Q(Y, a')) \\ &= (1 - 0.5) \times 0 + 0.5 \times (0 + 0.5 \times 0) \\ &= 0 \end{aligned}$$

At $t = 3$, we get:

$$\begin{aligned} Q(Y, Right) &= (1 - \alpha)Q(Y, Right) + \alpha(r + \gamma \max_{a'} Q(X, a')) \\ &= (1 - 0.5) \times 0 + 0.5 \times (3 + 0.5 \times 1) \\ &= 1.75 \end{aligned}$$

At $t = 4$, we get:

$$\begin{aligned} Q(X, Right) &= (1 - \alpha)Q(X, Right) + \alpha(r + \gamma \max_{a'} Q(X, a')) \\ &= (1 - 0.5) \times 0 + 0.5 \times (-1 + 0.5 \times 1) \\ &= -0.25 \end{aligned}$$

So the final values are:

$$Q(X, Right) = -0.25$$

$$Q(X, Left) = 1$$

$$Q(Y, Right) = 1.75$$

$$Q(Y, Left) = -2$$

Rubric:

- 4 points for each iteration.
- 1 point for each value.

b) Determine the optimal policy.

6
(CO3)
(PO3)

Solution:

$$\begin{aligned} \pi(X) &= \operatorname{argmax} (Q(X, Right), Q(X, Left)) \\ &= Left \end{aligned}$$

$$\begin{aligned} \pi(Y) &= \operatorname{argmax} (Q(Y, Right), Q(Y, Left)) \\ &= Right \end{aligned}$$

Rubric:

- 3 points for each correct policy.

2. Consider creating a chatbot application where users ask questions, the chatbot replies, and users rate the answers. The rating dictates the utility of the chatbot. If the chatbot is unsure about an answer, it can choose actions as specified in Table 2.

Table 2: Possible chatbot scenarios for Question 2

Action	Potential Outcome	Probability	Utility
Say "I don't know"	User gets bored	1.0	0
Guess the answer	Correct answer	0.6	1
	Wrong answer	0.4	X
Trick the user to click "like"	Tricking succeeds	p	1
	Tricking fails	$1 - p$	$2X$

- a) Assume $X = -1$ and $p = 0$. With proper justification, recommend the optimal action for a rational chatbot.

10
(CO3)
(PO3)

Solution:

$$U(IDK) = 0$$

$$U(Guess) = 0.6 \times 1 + 0.4 \times (-1) = 0.2$$

$$U(Trick) = 0 \times 1 + 1 \times 2 \times (-1) = -2$$

The optimal action is to guess the answer since it has the highest utility.

Rubric:

- 2 points for the correct recommendation
- 2 point each for utility
- 2 point for justification

- b) Assume that $p = 0.7$. Determine the necessary and sufficient condition in terms of X so that a rational chatbot performs the same action as recommended in Question 2(a).

8
(CO1)
(PO1)

Solution:

We need:

$$U(Guess) > U(Trick)$$

$$0.6 \times 1 + 0.4 \times X > 0.7 \times 1 + 0.3 \times 2 \times X$$

$$X < -1/2$$

Again:

$$U(Guess) > U(IDK)$$

$$0.6 \times 1 + 0.4 \times X > 0$$

$$X > -3/2$$

So the necessary and sufficient condition is $X \in \left(-\frac{3}{2}, -\frac{1}{2}\right)$

Rubric:

- 3 points for each calculation
- 1 point for each correct bound

3. In Figure 1, a simplified map shows seven cities connected by bidirectional roads, each associated with a positive integer travel time. A robot plans to journey from city S to city T using Graph Search variant of search algorithms to find the path.

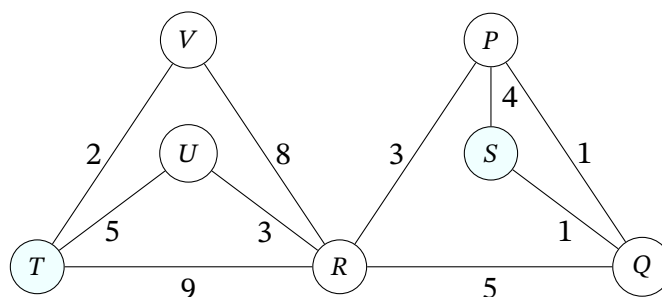


Figure 1: A simplified map of a country for Question 3

Two heuristics h_0 and h_1 are shown in Table 3. Here, h_0 is consistent, but h_1 is not consistent.

Following is a list of paths which can be possibly be returned by a search algorithm:

Table 3: Heuristic Functions for Question 3

Node	P	Q	R	S	T	U	V
h_0	8	9	7	9.5	0	4	1.5
h_1	10	12	8	10	0	4.5	1

- **Path 1:** $S \rightarrow P \rightarrow R \rightarrow T$
- **Path 2:** $S \rightarrow Q \rightarrow R \rightarrow T$
- **Path 3:** $S \rightarrow Q \rightarrow P \rightarrow R \rightarrow U \rightarrow T$

- a) For each of the following algorithms, state the path(s) (among the aforementioned paths) returned: 5 × 6
(CO1)
(PO1)
- i. Depth First Search
 - ii. Breadth First Search
 - iii. Uniform Cost Search
 - iv. A* Search with Heuristic h_0
 - v. A* Search with Heuristic h_1

Solution:

- i. Path 1, 2, 3
- ii. Path 1, 2
- iii. Path 3
- iv. Path 3
- v. Path 3

Rubric:

- 2 points for each correctly identified path
- 2 points for each correctly omitted path

- b) You are designing a new heuristic function h_2 for the given scenario. You have come up with all the values except $h_2(Q)$ as shown in Table 4: 3 × 10
(CO3)
(PO3)

Table 4: Incomplete Heuristic Functions for Question 3

Node	P	Q	R	S	T	U	V
h_2	9	?	7	10	0	4.5	1.5

For each of the following scenarios, write the possible range of values for $h_2(Q)$ with a brief explanation:

- i. h_2 will be admissible.

Solution:

To make h_2 admissible, $h_2(Q)$ has to be less than or equal to the actual optimal cost from Q to goal T , which is the cost of the path $Q \rightarrow P \rightarrow R \rightarrow U \rightarrow T$, i.e. 12. And the minimum possible value for heuristic is no heuristic, which makes all the values 0.

The answer is $0 \leq h_2(Q) \leq 12$.

Rubric:

- 2 points for each correct range
- 3 point for correct explanation of each bound.

ii. h_2 will be consistent.

Solution:

All the other nodes except node Q satisfy the consistency conditions. The consistency conditions that involve the state Q are:

- | | |
|---|--|
| • $h(S) \leq c(S, Q) + h(Q)$
$\rightarrow 10 \leq 1 + h(Q)$
$\rightarrow 9 \leq h(Q)$ | • $h(Q) \leq c(Q, S) + h(S)$
$\rightarrow h(Q) \leq 1 + 10$
$\rightarrow h(Q) \leq 11$ |
| • $h(P) \leq c(P, Q) + h(Q)$
$\rightarrow 9 \leq 1 + h(Q)$
$\rightarrow 8 \leq h(Q)$ | • $h(Q) \leq c(Q, P) + h(P)$
$\rightarrow h(Q) \leq 1 + 9$
$\rightarrow h(Q) \leq 10$ |
| • $h(R) \leq c(R, Q) + h(Q)$
$\rightarrow 7 \leq 5 + h(Q)$
$\rightarrow 2 \leq h(Q)$ | • $h(Q) \leq c(Q, R) + h(R)$
$\rightarrow h(Q) \leq 5 + 7$
$\rightarrow h(Q) \leq 12$ |

Using the tightest bounds from the equations above: $9 \leq h_2(Q) \leq 10$.

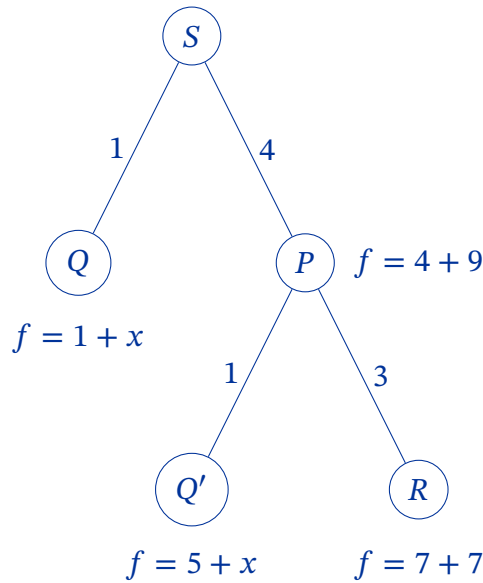
Rubric:

- 2 points for each correct range
- 3 points for correct explanation of each bound.

iii. A* Graph Search will traverse cities in the order: $S \rightarrow P \rightarrow Q \rightarrow R$.

Solution:

Let, $h_2(Q) = x$.



The A* search tree using the heuristic h_2 is shown above. Node S is expanded first since it's the start state.

To make A* graph search expand node P before node Q , we need

$$1 + x > 13$$

$$x > 12$$

After expanding P , we need to either expand Q' or Q before R . To expand Q' ,

$$5 + x < 14$$

$$x < 9$$

which violates the earlier constraint. To expand Q ,

$$1 + x < 14$$

$$x < 13$$

So we can get $12 < h_2(Q) < 13$

Rubric:

- 2 points for each correct range
- 3 points for correct explanation of each bound.

4. Consider that an agent is moving in the Gridworld shown in Figure 2. The only action available in the cells A and E is `Exit` with rewards 0 and X , respectively. Here, X = The last digit of your student ID + 1. For the other cells, the agent can take the action `Left` or `Right`, which results in the agent moving to the immediate left or right cell, respectively. There is no living reward or penalty. All actions succeed with probability 1. The discount factor is 1.

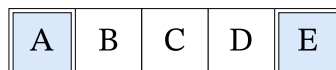


Figure 2: Gridworld for Question 4

Assume that we will run policy iteration to find the optimal action for the agent. In case of ties, we will choose `Left` instead of `Right`. The initial policy, $\pi_0(s) = \text{Left}$ for $s \in \{B, C, D\}$ and $\pi_0(s) = \text{Exit}$ for $s \in \{A, E\}$.

- a) Formulate the scenario as a Markov Decision Process.

Solution:

- States, $S = \{A, B, C, D, E\}$
- Actions, $A = \{\text{Left}, \text{Right}, \text{Exit}\}$
- Terminal States: $\{A, E\}$

Rubric:

- 1 point for each item.

b) With a brief explanation, determine the resulting policy after one iteration.

19
(CO1)
(PO1)

Solution:

We first run policy evaluation to learn the value of the current policy, π_0 . Given the expected discounted reward for going left is 0, we have $V^{\pi_0}(B) = V^{\pi_0}(C) = V^{\pi_0}(D) = 0$. We also have $V^{\pi_0}(A) = 0$ and $V^{\pi_0}(E) = X$ since the only actions available from those two states are to `Exit` and collect the reward.

Then, we run a one-step look-ahead to compute the new policy, π_1 . The general equation for policy improvement is: $\pi_{i+1}(s) = \operatorname{argmax}_a T(s, a, s') [R(s, a, s') + \gamma V^{\pi_i}(s')]$.

However, in this question, we have deterministic actions, so the expected value computation over s' is not required. We can also plug in $\gamma = 1$. At the same time, since the living reward is always 0, we can ignore the $R(s, a, s')$ term as well for non-exit actions. So the simplified equation is: $\pi_{i+1}(s) = \operatorname{argmax}_a [V^{\pi_i}(\text{state left of } s), V^{\pi_i}(\text{state right of } s)]$.

Plugging in the values for each state:

$$\pi_1(B) = \operatorname{argmax}_a [V^{\pi_0}(A), V^{\pi_0}(C)] = \operatorname{argmax}[0, 0] = \text{Left}.$$

$$\pi_1(C) = \operatorname{argmax}_a [V^{\pi_0}(B), V^{\pi_0}(D)] = \operatorname{argmax}[0, 0] = \text{Left}.$$

$$\pi_1(D) = \operatorname{argmax}_a [V^{\pi_0}(C), V^{\pi_0}(E)] = \operatorname{argmax}[0, X] = \text{Right}.$$

$\pi_1(A)$ and $\pi_1(E)$ remains the same.

Rubric:

- 1 point for each correct policy evaluation.
- 2 point for each correct policy improvement.
- 2 points for each explanation.

c) Given that we start with the policy π_0 , determine the number of iterations of policy iteration required to compute the optimal policy.

2
(CO1)
(PO1)

Solution:

Total 3 iterations are required. The first iteration changes the policy at D . The next iteration changes the policy at C . Then, the third and final iteration changes the policy at B .

Rubric:

- 2 points for correct answer.

5. After the yearly checkup of a patient, an AI-based agent, Alice said it has a ‘bad’ news and a ‘good’ news. The bad news is that the patient tested positive for a serious disease and that the test is 99% accurate. The good news is that this is a rare disease, striking only 1 in 10,000 people of the patient’s age. Another AI-based agent, Bob concluded that the patient actually has the disease. 5 + 6
(CO3)
(PO3)

- a) Provide arguments to support the decision of Alice labeling the good news as ‘good’.

Solution:

We are given the following information:

$$P(\text{test}|\text{disease}) = 0.99$$

$$P(\neg\text{test}|\neg\text{disease}) = 0.99$$

$$P(\text{disease}) = 0.0001$$

and the observation *test*. What the patient is concerned about is $P(\text{disease}|\text{test})$. Roughly speaking, the reason it is a good thing that the disease is rare is that $P(\text{disease}|\text{test})$ is proportional to $P(\text{disease})$, so a lower prior for *disease* will mean a lower value for $P(\text{disease}|\text{test})$. Roughly speaking, if 10,000 people take the test, we expect 1 to actually have the disease, and mostly likely test positive, while the rest do not have the disease, but 1% of them (about 100 people) will test positive anyway, so $P(\text{disease}|\text{test})$ will be about 1 in 100.

Rubric:

- 5 points for correct argument

- b) Criticize the decision made by Bob.

Solution:

We know that,

$$\begin{aligned} P(\text{disease}|\text{test}) &= \frac{P(\text{test}|\text{disease})P(\text{disease})}{P(\text{test}|\text{disease})P(\text{disease}) + P(\text{test}|\neg\text{disease})P(\neg\text{disease})} \\ &= \frac{0.99 \times 0.0001}{0.99 \times 0.0001 + 0.01 \times 0.9999} \\ &= 0.009804 \end{aligned}$$

The likelihood of Bob’s decision being correct is 0.9804%. So it is highly likely that Bob is wrong.

Rubric:

- 3 points for the decision
- 3 points for the calculation

Appendix

- $V^*(s) = \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V^*(s')]$
- $V_{k+1}(s) = \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V_k(s')]$, where $V_0(s) = 0$
- $V^\pi(s) = \sum_{s'} T(s, \pi(s), s') [R(s, \pi(s), s') + \gamma V^\pi(s')]$
- $V_{k+1}^\pi(s) = \sum_{s'} T(s, \pi(s), s') [R(s, \pi(s), s') + \gamma V_k^\pi(s')]$, where $V_0^\pi(s) = 0$
- $\pi^*(s) = \operatorname{argmax}_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V^*(s')]$
- $\pi_{i+1}(s) = \operatorname{argmax}_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V^{\pi_i}(s')]$
- $\pi^*(s) = \operatorname{argmax}_a Q^*(s, a)$
- $V^\pi(s) = (1 - \alpha)V^\pi(s) + \alpha \times \text{sample}$, where $\text{sample} = R(s, \pi(s), s') + \gamma V^\pi(s')$
- $Q_{k+1}(s, a) = \sum_{s'} T(s, a, s') \left[R(s, a, s') + \gamma \max_{a'} Q_k(s', a') \right]$
- $Q(s, a) = R(s, a, s') + \gamma \max_{a'} f(Q(s', a'), N(s', a'))$
- $w_i = w_i + \alpha[\text{difference}]f_i(s, a)$, where $\text{difference} = \left[r + \gamma \max_{a'} Q(s', a') \right] - \hat{Q}(s, a)$
- $P(x|y) = \frac{P(x,y)}{P(y)}$
- $P(x_1, x_2, \dots, x_n) = \prod_i P(x_i|x_1, \dots, x_{i-1})$
- $P(x|y) = \frac{P(y|x)}{P(y)}P(x)$